Independent *t* Test

ITAO 20200: Foundations of Statistics

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Description

An independent t test compares the means of two independent, unrelated groups.

Assumptions

The independent variable (i.e., the factor, or the grouping variable) should only have two levels, and the levels should be independent of one another. There is *no* matching or pairing of participants in the two groups. Dependent variables should be measured on interval or ratio scales. The scores should be normally distributed, but the t test is robust to this assumption and can still be performed if this assumption is violated. The homogeneity of variance assumption can be tested for in R.

Data Entry

For this test, two separate variable columns must be created: one for the dependent variable, and one for the factor. For this test, we will use the data file `BankSalaries.csv`. The `Salary` variable measures employee salaries at a bank. The `Gender` variable indicates how an employee is classified (0 = Male, 1 = Female). The data set will be read into R and called `bank`.

```
bank <- read.csv("https://www.crc.nd.edu/~jwaddell/ITAO20200/Data/BankSalaries.csv")
head(bank, n = 5) # First 5 observations
```

```
##   EmployeeNumber      EducLev JobGrade YrsExper Age Gender YrsPrior IT_Job Salary
## 1             95    Some Grad        2        2  27      0        2     No  30000
## 2             38 Some College        1        2  30      1        4     No  28500
## 3             50 Some College        1        2  53      1        0     No  30450
## 4             22 Some College        1        3  28      0        3     No  31000
## 5             24 Some College        1        3  33      0        4     No  26700
```

First, let's transform our nominal grouping variable, `Gender`, into a factor. This will recode the 0s and 1s into Male and Female, respectively.

```
bank$Gender <- factor(bank$Gender, levels = c(0, 1),
                      labels = c("Male", "Female"))
```

After the transformation, this is how our data should look:

```
head(bank, n = 5) # First 5 observations
```

```
##   EmployeeNumber      EducLev JobGrade YrsExper Age Gender YrsPrior IT_Job Salary
## 1             95    Some Grad        2        2  27   Male        2     No  30000
## 2             38 Some College        1        2  30 Female        4     No  28500
## 3             50 Some College        1        2  53 Female        0     No  30450
## 4             22 Some College        1        3  28   Male        3     No  31000
## 5             24 Some College        1        3  33   Male        4     No  26700
```

Data Analysis and Output

Our objective is to determine if the means of the groups differ significantly from one another; if so, we can conclude that the sample means represent two different populations.

Summary Statistics

Since our analysis focuses on group differences, it is important that the summary statistics reflect values for each group. One way to separate salary values based on gender groups is to use the `by` command. Individual commands can be run for specific summary statistics, such as group means, standard deviations, and sample sizes.

```
by(bank$Salary, bank$Gender, mean)  #Dependent var, Independent var, statistic
```

```
## bank$Gender: Male
## [1] 45505.44
## ------------------------------------------------------------
## bank$Gender: Female
## [1] 37209.93
```

```
by(bank$Salary, bank$Gender, sd)  #Dependent var, Independent var, statistic
```

```
## bank$Gender: Male
## [1] 15843.22
## ------------------------------------------------------------
## bank$Gender: Female
## [1] 6710.867
```

```
by(bank$Salary, bank$Gender, length)  #Dependent var, Independent var, statistic
```

```
## bank$Gender: Male
## [1] 68
## ------------------------------------------------------------
## bank$Gender: Female
## [1] 140
```

More Advanced Code for Summary Statistics Rather than run individual lines of code reflecting specific descriptive statistics, some might find it preferable to write one block of code that calculates numerous statistics simultaneously. The following block of code calculates a set of summary statistics for employee salaries, broken down separately by gender. The `by()` function is still the key command here: it

splits the Salary column into groups based on the categories in the Gender column (Male vs. Female), and then applies a custom calculation to each group separately.

The custom calculation is defined inside the `function(x)` part of the code, where x stands in for the salary values within each gender group. For each group, it computes seven statistics: the number of people in that group (n), the average salary (mean), the standard deviation (sd), the median, the minimum and maximum salary in the group (min and max), and the standard error of the mean (se). Most of these values are rounded to two decimal places for readability.

The end result is a side-by-side comparison; for each gender, you get a compact summary of how many employees there are, what their typical salary looks like, how much salaries vary, and the range of salaries observed. This makes it easy to compare pay patterns between groups at a glance.

```
by(bank$Salary, bank$Gender, function(x) {
  c(n      = length(x),
    mean   = round(mean(x), 2),
    sd     = round(sd(x), 2),
    median = round(median(x), 2),
    min    = min(x),
    max    = max(x),
    se     = round(sd(x) / sqrt(length(x)), 2))
})
```

```
## bank$Gender: Male
##        n     mean       sd   median      min      max       se
##    68.00 45505.44 15843.22 42500.00 26700.00 97000.00  1921.27
## ------------------------------------------------------------------------
## bank$Gender: Female
##        n     mean       sd   median      min      max       se
##   140.00 37209.93  6710.87 35450.00 26800.00 61800.00   567.17
```

Test for Homogeneity (Equality) of Variance

Before proceeding to the independent t test, we need to determine if the variances of our male and female salary values are equal or unequal to one another. This impacts the type of independent t test we perform.

We will use Levene's Test to make this determination. The `leveneTest()` function can be found in the `car` package. (**Note:** if the package is not already installed on your machine, look in the "Packages" tab in the bottom right-hand portion of the window, click on "Install", type in `car`, and then click on 'Install", or run `install.packages("car")`.)

First, the package must be called with the `library` command. Then, certain variables must be identified within the statement itself. The first argument, corresponding to `y =` below, identifies the dependent variable. In our case, that is the Salary data. The second argument, corresponding to `group =` below, identifies the factor, or independent variable. In this case, that is Gender. The final argument, `center = mean`, tells R to use the group means as the reference point when checking for differences in group variability estimates.

```
library(car)
leveneTest(y = bank$Salary,        # dependent variable
           group = bank$Gender,    # independent variable
           center = mean)          # for Levene's test
```

```
## Levene's Test for Homogeneity of Variance (center = mean)
##        Df F value    Pr(>F)
## group   1  32.057 4.983e-08 ***
##       206
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The results provide the probability that the two sample **variances** are statistically similar to one another at the population level. Levene's test is an F test; hence, it has a one-tailed $p-$value. If the $p-$value in the above output (denoted as `Pr(>F)`) is greater than 0.05, we **assume equal variances** for our independent t test. However, if the $p-$value in the above output is less than or equal to 0.05, we will **not assume equal variances** (or, in other words, perform an unequal variances assumed test). *Note:* Our textbook teaches us to perform the independent t using the unequal variances approach.

With this information, let's proceed with our independent t test. Though our Levene's test shows unequal variances should be assumed, in the following, we will perform an independent t test using both approaches for comparison purposes.

The hypotheses of interest here are:

$$H_0 : \mu_1 - \mu_2 = 0$$

$$H_a : \mu_1 - \mu_2 \neq 0$$

To test our hypotheses, we will use the `t.test()` function in R. The `alternative` argument allows us to set the type of alternative hypothesis we have. For a two-tailed test, use `alternative = two.sided`. For a one-tail lower-tail test, use `alternative = less`. For a one-tail upper-tail test, use `alternative = greater`.

The analysis is set to run with a default setting of $\alpha = 0.05$. If you wish to change this, modify the `conf.level =` line in the `t.test()` function. The `var.equal =` option is used to specify whether equal variances should be assumed. When set to `var.equal = TRUE`, equal variances are assumed; when set to `var.equal = FALSE`, equal variances are not assumed.

Output 1 : Unequal Variances Assumed

```
# Independent t test for analyzing the differences between Salary of Male
# and Female employees when unequal variances are assumed

t.test(bank$Salary ~ bank$Gender,     # variables tested (depen. ~ indep.)
       alternative = "two.sided",     # specify the type of alternative hypothesis
       mu = 0,                        # identifies the null value
       conf.level = 0.95,             # confidence level = 1 - alpha
       var.equal = FALSE)             # FALSE = **assumes unequal variances**
```

```
##
##  Welch Two Sample t-test
##
## data:  bank$Salary by bank$Gender
## t = 4.141, df = 78.898, p-value = 8.604e-05
## alternative hypothesis: true difference in means between group Male and group Female
↪  is not equal to 0
## 95 percent confidence interval:
##   4308.082 12282.943
## sample estimates:
##   mean in group Male mean in group Female
##             45505.44             37209.93
```

Output 2 : Equal Variances Assumed

```
# Independent t test for analyzing the differences between Salary of Male
# and Female employees when equal variances are assumed

t.test(bank$Salary ~ bank$Gender,     # variables tested (depen. ~ indep.)
       alternative = "two.sided",     # specify the type of alternative hypothesis
       mu = 0,                        # the null (or test) value
       conf.level = 0.95,             # confidence level = 1 - alpha
       var.equal = TRUE)              # TRUE  = **assumes equal variances**
```

```
##
##  Two Sample t-test
##
## data:  bank$Salary by bank$Gender
## t = 5.3024, df = 206, p-value = 2.935e-07
## alternative hypothesis: true difference in means between group Male and group Female
↪  is not equal to 0
## 95 percent confidence interval:
##   5211.041 11379.984
## sample estimates:
```

```
##   mean in group Male mean in group Female
##             45505.44             37209.93
```

The output for the independent t test when unequal variances are assumed is labeled **Welch Two Sample t-test**, and the output is labeled **Two Sample t-test** when equal variances are assumed.

Both outputs give us the obtained test statistic, the degrees of freedom, the p-value, information about the type of alternative hypothesis requested in the code, the 95% confidence interval of the true difference between means, and the sample mean estimates in each group.

Note that the t values and degrees of freedom (df) are calculated differently depending on whether equal variances are assumed or not. These differences contribute to the differing p-values as well, which indicate the exact probability of our obtained test statistic as a two-tailed test.

The *95 percent confidence interval* is built around the difference between the means (as calculated by average male salary - average female salary, according to group assignments); it indicates the range of mean differences that we believe could represent the true mean difference between male and female salaries.

Interpreting the Output

If the p-value of the test statistic (using the correct output) is less than or equal to your alpha, then there is a statistically significant difference between the group means. This suggests that there is a very low probability of obtaining such an extreme difference if, in fact, the null hypothesis is true. If the p-value reported is greater than your alpha, then your results are inconclusive.

You can also determine statistical significance by evaluating the confidence interval results. To make this decision, look for the hypothesized null value (0). If 0 is inside the confidence interval, retain the null and conclude that no significant difference exists between the group means. If 0 is outside the confidence interval, reject the null of no difference in favor of the alternative.

Writing Your Results

To convey your results properly, you need to know the t value, the degrees of freedom, and the $p-$value of the test statistic. Additionally, it would be helpful to have the mean and standard deviation of the groups. When writing the results of an unequal variances t test, the degrees of freedom should be rounded down to a whole integer.

In this example, there is a statistically significant difference between the average male and female salaries. To determine *how* they are different, return to the sample means and make an evaluative statement. In this situation, the average male salary ($45,505) is significantly higher than the average female salary ($37,210). We can relate this finding with something like this:

> An independent t test was performed to determine if salary differences exist between male and female employees. The results indicated that salary does differ significantly by the gender of the employee, $t(78) = 4.141,\ p = 0.0000$. Male employees ($\bar{x} = 45{,}505,\ s = 15{,}843$) have a significantly higher average salary than female employees ($\bar{x} = 37{,}210,\ s = 6{,}711$).

Practice on Your Own

FedEx and United Parcel Service (UPS) are the world's two leading cargo carriers by volume and revenue. According to the Airports Council International, the Memphis International Airport (FedEx) and the Louisville International Airport (UPS) are 2 of the 10 largest cargo airports in the world. The data set contains tons of cargo per day handled by these airports. Data are in thousands of tons. Can you determine if FedEx handles a significantly different amount of cargo than UPS on a given day at these airports? Use an alpha of 0.05. (Data set on Canvas called `Cargo.csv`; 1 = FedEx, 2 = UPS. First download the data set, and then import it into R using the instructions found in Importing Data Sets in R.)

1. What are the appropriate null and alternative hypotheses for this situation? Complete the remaining portion.

$$H_0 : \mu_1 - \mu_2 _____$$

$$H_a : \mu_1 - \mu_2 _____$$

2. Mean for Fed Ex: _____ Mean for UPS:_____
3. Based on the output, should we assume equal variances? How did you come to that conclusion?
4. Report the appropriate t-value.
5. We know the reported difference between means is unlikely to be the true difference between the population means of how much cargo Fed Ex and UPS move on a given day. Between what values do we think the true difference between means is found?
6. Based on the evidence, should we reject or retain the null?
7. Write the results in proper statistical notation.
8. TRUE or FALSE: We can conclude that Fed Ex and UPS move different amounts of cargo each day (on average). Fed Ex seems to move more cargo than UPS.